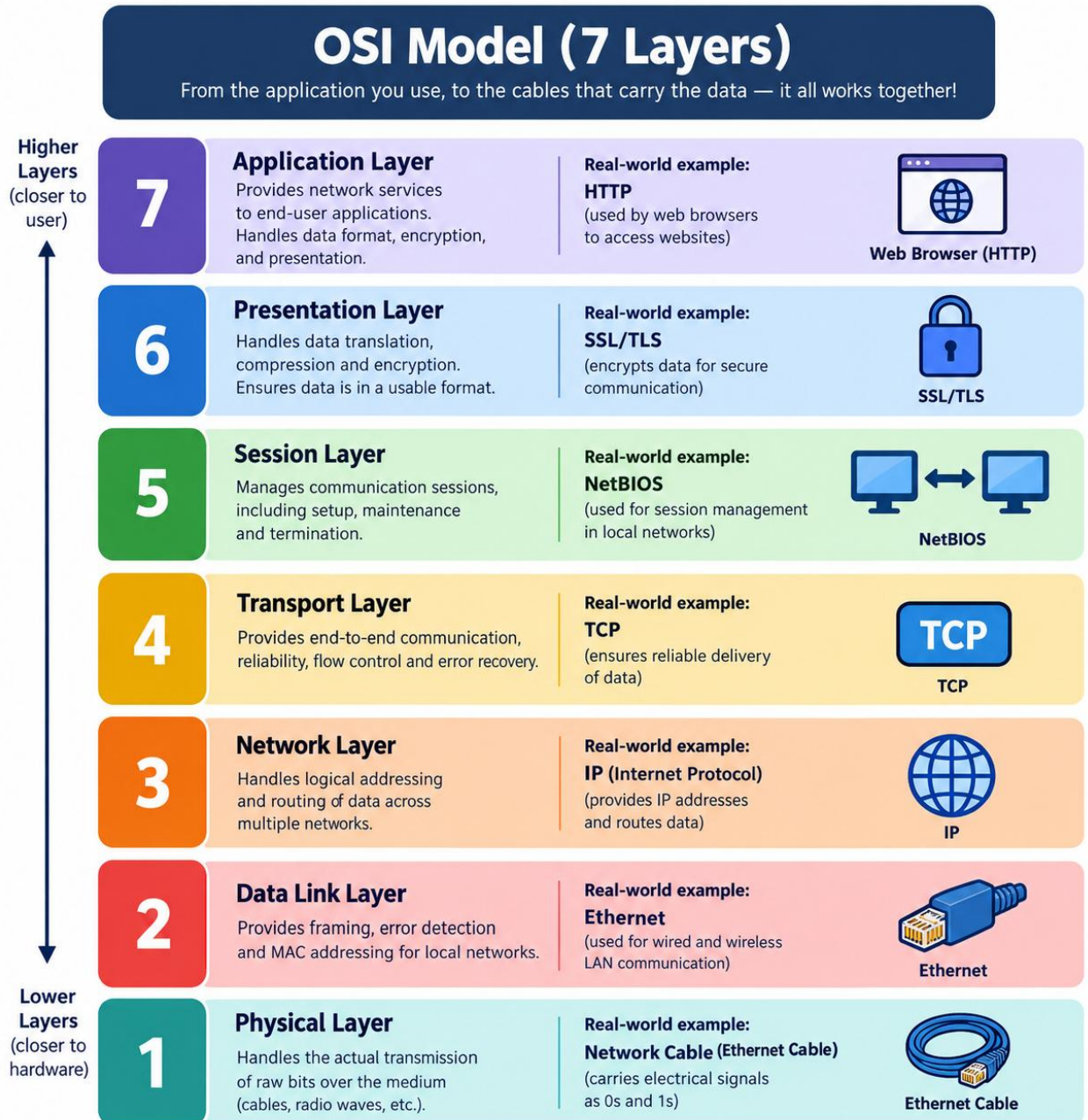


INTRODUCTION TO NETWORKING AND THE OSI MODEL

1. Draw an OSI Model diagram showing all 7 layers, and for each layer, write one real-world example (such as HTTP for Application, TCP for Transport, Ethernet for Data Link, cables for Physical, etc).

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2. List three apps you use daily (for example: Instagram, Zomato, or WhatsApp) and for each, pick one network activity (like sending a message or placing an order) and explain which OSI layers are involved in that activity.

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App	Network activity	OSI layers involved
WhatsApp	Sending a message	7 Application: WhatsApp messaging service → 6 Presentation: encryption/formatting → 5 Session: maintains the connection → 4 Transport: TCP/UDP carries data → 3 Network: IP routes the message → 2 Data Link: Wi-Fi/Ethernet transfers frames → 1 Physical: radio waves/cables transmit bits
Instagram	Loading a photo	7 Application: Instagram/HTTP or HTTPS → 6 Presentation: data encoding & encryption → 5 Session: manages the communication session → 4 Transport: TCP/QUIC transports data → 3 Network: IP routing → 2 Data Link: Wi-Fi/mobile network frames → 1 Physical: radio waves or cables carry the bits
Zomato	Placing a food order	7 Application: Zomato's ordering service → 6 Presentation: encryption and data formatting → 5 Session: maintains the connection → 4 Transport: transports order data → 3 Network: IP routes the request to Zomato's servers → 2 Data Link: Wi-Fi/mobile network handles local delivery → 1 Physical: radio signals/cables transmit the actual bits

In simple terms

Think of the OSI model as **7 steps for moving information:**

Application → Presentation → Session → Transport → Network → Data Link → Physical

For example, when you send a WhatsApp message, the message starts at the **Application layer**, gets formatted and protected, transported across networks, routed using **IP**, converted into network frames, and finally transmitted as **electrical/radio signals** at the Physical layer.

3. Install and open Wireshark on your computer, start a capture, then visit any website (for example, www.example.com) in your browser. Stop the capture and save the PCAP file.

-> I can't directly install and run Wireshark on your computer from this chat, but you can complete the task as follows:

1. Download and install **Wireshark** from the official site: [Wireshark Download](#)
2. Open **Wireshark**.
3. Select your active network interface:
 - **Wi-Fi** if you're connected wirelessly.
 - **Ethernet** if you're using a cable.
4. Click the **Start Capturing Packets** button (shark-fin icon).
5. Open your browser and visit <https://www.example.com>.
6. Wait a few seconds for the page to load.
7. Return to Wireshark and click the **Stop** button.
8. Go to **File → Save As**.
9. Save the capture as, for example:
website_capture.pcapng
10. Your saved file can then be submitted as the **PCAP capture**.

Tip: .pcapng is Wireshark's modern default format; if your assignment specifically requires .pcap, select that format when saving.

4. In Wireshark, find a captured HTTP request packet from your browsing session, take a screenshot of the packet details, and annotate which parts of the packet correspond to the Application, Transport, Network, and Data Link layers.

Hint: Use the packet details pane in Wireshark to expand each protocol section (HTTP, TCP, IP, Ethernet).

-> You can complete this directly in Wireshark. Here's exactly what to do:

1. Find an HTTP packet

Open your saved capture in Wireshark and enter this filter:

http.request

Press **Enter**. Look for a packet showing something such as:

GET / HTTP/1.1

Host: example.com

If nothing appears, your browser may have used HTTPS. Try browsing to a site that supports plain HTTP or capture a new session where HTTP traffic is available.

2. Expand the packet details

Click the HTTP request packet, then expand these sections in the **Packet Details** pane:

- ▼ Ethernet II
- ▼ Internet Protocol Version 4
- ▼ Transmission Control Protocol
- ▼ Hypertext Transfer Protocol

3. Annotate the OSI layers

OSI Layer	Wireshark section	What to point out
Application – Layer 7	HTTP	GET, Host, HTTP version, requested URL/path
Transport – Layer 4	TCP	Source/destination ports, sequence number, acknowledgment
Network – Layer 3	IPv4	Source IP and destination IP addresses
Data Link – Layer 2	Ethernet II	Source and destination MAC addresses

For your assignment: take a screenshot with the four protocol sections expanded and add arrows/text labels identifying **Layer 2 – Ethernet, Layer 3 – IP, Layer 4 – TCP, and Layer 7 – HTTP**.

If you **upload your Wireshark screenshot here**, I can annotate the actual screenshot for you.

